CSIS1614 Mixed Revision Test - LU3, LU4, LU5, LU6

60 questions total - 15 questions per learning unit. Difficulty increases as each unit progresses. Use this as a written revision paper. Questions are different from the earlier practice questions.

LU3 - Making Decisions

Boolean data type and expressions

1. State the only two values that a bool variable can store.

2. Classify the operator != as equality, relational, or logical.

3. Classify the operator ! as equality, relational, or logical.

One-way and two-way selection

4. What will this print?

```
if (12 < 20)
{
    Console.WriteLine("A");
}
```

5. What will this print?

```
int x = 9;
if (x % 2 == 0)
{
    Console.WriteLine("Even");
}
else
{
    Console.WriteLine("Odd");
}
```

6. Complete the sentence:

```
A one-way if statement executes its body only when the condition is ______.
```

Nested if and logical operators

7. What will this print?

```
int score = 72;
if (score >= 50)
{
    if (score >= 75)
        Console.WriteLine("Distinction");
    else
        Console.WriteLine("Pass");
}
else
{
    Console.WriteLine("Fail");
}
```

8. What will this print?

```
int age = 17;
bool hasCard = true;
if (age >= 18 || hasCard)
{
    Console.WriteLine("Enter");
}
else
{
    Console.WriteLine("Wait");
}
```

9. What will this print?

```
int a = 8;
if (a > 5 && a < 8)
{
    Console.WriteLine("Inside");
}
else
{
    Console.WriteLine("Outside");
}
```

switch and ternary operator

10. What will this print?

```
char grade = 'B';
switch (grade)
{
    case 'A':
        Console.WriteLine("Excellent");
        break;
    case 'B':
        Console.WriteLine("Good");
        break;
    default:
        Console.WriteLine("Try again");
        break;
}
```

11. Rewrite the idea below using the ternary operator:

```
If mark is at least 50, result must be "Pass"; otherwise result must be "Fail".
```

12. What will this print?

```
int temp = 30;
string msg = temp > 25 ? "Hot" : "Cold";
Console.WriteLine(msg);
```

Short-circuit evaluation and scope

13. In the condition below, if x > 100 is false, will C# still evaluate y < 0?

```
if (x > 100 && y < 0)
```

14. In the condition below, if ready is true, will C# still evaluate count > 5?

```
if (ready || count > 5)
```

15. State whether this is valid or invalid, and briefly say why:

```
if (true)
{
    int n = 4;
}
Console.WriteLine(n);
```

LU4 - Repeating Instructions

Loop purpose and components

16. State the main purpose of a loop in a program.

17. For the loop below, identify the control variable, condition, and update statement:

```
int i = 2;
while (i <= 8)
{
    Console.WriteLine(i);
    i += 2;
}
```

18. Name the two broad categories of loops based on when the condition is tested.

while, for, do...while

19. What will this print?

```
int i = 3;
while (i < 6)
{
    Console.WriteLine(i);
    i++;
}
```

20. What will this print?

```
for (int i = 5; i >= 3; i--)
{
    Console.WriteLine(i);
}
```

21. What will this print?

```
int i = 7;
do
{
    Console.WriteLine(i);
} while (i < 5);
```

Loop control types

22. A program asks the user to enter quantities and uses 0 to stop input. What loop-control type is this?

23. A loop continues while a bool variable named keepGoing remains true. What loop-control type is this?

24. A loop runs from 1 to 12 using for (int m = 1; m <= 12; m++). What loop-control type is this?

break and continue

25. What will this print?

```
for (int i = 1; i <= 6; i++)
{
    if (i == 5)
        break;
    Console.Write(i);
}
```

26. What will this print?

```
for (int i = 1; i <= 6; i++)
{
    if (i % 2 == 0)
        continue;
    Console.Write(i);
}
```

Nested loops

27. How many times will Console.Write("X") run?

```
for (int row = 1; row <= 4; row++)
{
    for (int col = 1; col <= 3; col++)
    {
        Console.Write("X");
    }
}
```

28. Write the full output:

```
for (int i = 1; i <= 3; i++)
{
    for (int j = 1; j <= i; j++)
    {
        Console.Write(i);
    }
    Console.WriteLine();
}
```

29. Will this pattern grow, shrink, or stay the same width?

```
for (int i = 1; i <= 4; i++)
{
    for (int j = 4; j >= i; j--)
    {
        Console.Write("*");
    }
    Console.WriteLine();
}
```

30. Write the full output:

```
for (int i = 1; i <= 2; i++)
{
    for (int j = 1; j <= 4; j++)
    {
        if (j == 3)
            continue;
        Console.Write(j);
    }
    Console.WriteLine();
}
```

LU5 - Methods and Behaviours

Method basics and execution flow

31. In the method below, identify the return type and method name:

```
static void ShowMenu()
{
    Console.WriteLine("Menu");
}
```

32. What is the method call in the code below?

```
static void Main(string[] args)
{
    PrintLine();
}
static void PrintLine()
{
    Console.WriteLine("----");
}
```

33. Write the output:

```
static void Main(string[] args)
{
    Console.WriteLine("A");
    Show();
    Console.WriteLine("B");
}
static void Show()
{
    Console.WriteLine("C");
}
```

Void vs value-returning methods

34. State whether this is a void method or a value-returning method:

```
static decimal GetPrice()
{
    return 19.95m;
}
```

35. What will be printed?

```
static void Main(string[] args)
{
    Console.WriteLine(GetNumber());
}
static int GetNumber()
{
    return 12;
}
```

Parameters and arguments

36. In this method header, what are the parameters?

```
static void Display(string name, int age)
{
    Console.WriteLine(name + " " + age);
}
```

37. In this call, what are the arguments?

```
Display("Lebo", 19);
```

38. What will be printed?

```
static void Main(string[] args)
{
```

```
    int total = Multiply(4, 3);
    Console.WriteLine(total);
}
static int Multiply(int x, int y)
{
    return x * y;
}
```

Value, ref, and out parameters

39. What will be printed?

```
static void Change(int x)
{
    x = 30;
}
static void Main(string[] args)
{
    int num = 7;
    Change(num);
    Console.WriteLine(num);
}
```

40. What will be printed?

```
static void Change(ref int x)
{
    x = x - 2;
}
static void Main(string[] args)
{
    int num = 9;
    Change(ref num);
    Console.WriteLine(num);
}
```

41. What will be printed?

```
static void Initialise(out int x)
{
    x = 44;
}
static void Main(string[] args)
{
    int value;
    Initialise(out value);
    Console.WriteLine(value);
}
```

Default, optional, and named parameters

42. What will be printed?

```
static void Main(string[] args)
{
    Welcome();
}
static void Welcome(string name = "Guest")
{
    Console.WriteLine("Welcome " + name);
}
```

43. What will be printed?

```
static void Main(string[] args)
{
    Welcome("Amo");
}
static void Welcome(string name = "Guest")
```

```
{
    Console.WriteLine("Welcome " + name);
}
```

44. What will be printed?

```
static void Main(string[] args)
{
    Show(second: 5, first: 2);
}
static void Show(int first, int second)
{
    Console.WriteLine(first + second);
}
```

45. State whether this call is valid or invalid:

```
Save(total: 100, name: "Neo");
for the method header
static void Save(string name, int total)
```

LU6 - Arrays and Strings

Array basics

46. How many elements does this array have?

```
int[] data = { 4, 8, 12, 16, 20 };
```

47. What is the last valid index of:

```
string[] words = new string[7];
```

48. What is the value of nums[3]?

```
int[] nums = { 2, 4, 6, 8, 10 };
```

Traversing arrays

49. Write the output:

```
int[] vals = { 3, 6, 9 };
for (int i = 0; i < vals.Length; i++)
{
    Console.WriteLine(vals[i]);
}
```

50. Write the output:

```
int[] vals = { 2, 5, 8 };
foreach (int v in vals)
{
    Console.WriteLine(v * 2);
}
```

51. Why is < preferred over <= in this loop header?

```
for (int i = 0; i < nums.Length; i++)
```

Common array patterns

52. What is the final total?

```
int[] nums = { 7, 14, 21 };
int total = 0;
foreach (int n in nums)
{
    total += n;
}
```

53. What is the highest value?

```
int[] nums = { 25, 40, 15, 55, 30 };
```

54. How many values are less than 30?

```
int[] nums = { 25, 40, 15, 55, 30, 10 };
```

Arrays with methods and references

55. State whether this method header is correct for receiving an array parameter:

```
static void PrintAll(int[] numbers)
```

56. What will be printed?

```
static void Main(string[] args)
{
    int[] nums = { 1, 2, 3 };
    Change(nums);
    Console.WriteLine(nums[1]);
}
static void Change(int[] nums)
{
    nums[1] = 99;
```

```
}
```

57. What will be printed?

```
int[] a = { 5, 6, 7 };
int[] b = a;
b[2] = 100;
Console.WriteLine(a[2]);
```

Strings as indexed collections and methods

58. What character is at index 4?

```
string word = "Table";
```

59. What does this return?

```
"College".IndexOf('l')
```

60. What does this return?

```
"Notebook".Remove(4, 2)
```